

Practical Data Science

Assignment 1: Data Preparation and Exploration

Student: Do Tung Lam – s3963286

Task 1 Data Preparation

1. Dataset Loading

The dataset *student_data_25s3.csv*, a modified subset of the UCI Student Performance dataset, was first loaded into the environment to establish a reliable foundation for analysis.

To preserve an immutable reference, the file was stored as **df_raw** (650 rows, 26 columns), which remains unchanged throughout data preparation. Because the dataset lacks a natural primary key and subsequent cleaning requires row-level tracking, a unique **row_id** column was added to every record. A separate working copy, **df = df_raw.copy()**, was then created to perform all cleaning and transformation steps while ensuring the original dataset is retained exactly as provided. [] image of loading

2. Initial Data Inspection

After importing the dataset, an initial profiling was conducted using `info()`, `describe()`, and `value_counts()` to assess data types, completeness, and adherence to the attribute definitions provided in the UCI Student Performance documentation. This step establishes whether the dataset conforms to the expected schema and identifies potential data quality issues requiring correction.

Data Type Assessment

Although the UCI documentation inconsistently labels *G1* and *G2* as categorical or target attributes, the dataset stores them as numeric grades, and *G3* is clearly the final target. Accordingly, **G1 and G2 are treated as numeric predictors**.

The expected numeric attributes—age, *G1*, *G2*, and the ordinal variable *studytime*—were incorrectly stored as object types. These mismatches typically arise from non-numeric tokens and prevent proper type coercion, obstructing valid statistical summaries and modelling.

Data Completeness

All current numeric type variables contain complete data across all 650 rows. Several object-type columns (**school**, **sex**, **Mjob**, **Fjob**, **studytime**, and **G1**)—show signs of potential missing values or mixed data types. These inconsistencies can disrupt categorical encoding and may introduce bias if the missingness mechanism is not random.

Data Validity

Comparing each feature against its defined domain revealed additional issues:

- failures contains values outside the valid 0–4 range.
- For object columns, address attribute should contain only 'U' (urban) or 'R' (rural), but 3 categories are observed, reason attribute is documented to include 4 categories, but 5 unique values were detected and more deviations columns will be logged in the next step.

These deviations indicate mislabeled or erroneous entries that compromise interpretability and model readiness. Given these findings, the next step is a systematic inspection of unique values in each categorical attribute to identify all incorrect, inconsistent, or undocumented categories.

3. Data Quality Audit

An audit of all current object-typed variables was conducted using `value_counts()`, `isna()` `duplicate()` and cross-referenced with the UCI dataset documentation.

Row Duplication: There are 6 duplicate rows found in the dataset. Because the dataset lacks a unique identifier, removing duplicates could risk discarding legitimate cases where different students share similar attributes. Therefore, the duplicates rows are not removed in the dataset.

Missing and Non-Standard Null Representations: The dataset contains only small percentage ($\leq 1\%$) of **both standard and hidden missing values**, often encoded as symbols or strings. Missing values appear in multiple non-standard forms, including 'X', 'nan', '0', and '-', rather than a unified null representation. Missing codes prevent correct null detection, interfere with categorical encoding, and may bias any downstream imputation procedures. Therefore, these missing required to be converted to a single NaN representation.

Incorrect Data Types and Mixed Formatting: Several variables that should contain integer values (e.g., age, G1, G2, studytime) were inferred as object type due to the presence of non-numeric values such as '1O', '2O', and 'X'. These typographical distortions violate the expected numeric domain and must be corrected to allow coercion into appropriate data types. Similarly, mixed formatting within the Mjob and Fjob attributes (e.g., 'at home', 'at_home', 'home') reflects semantic inconsistencies not supported by the documented categories.

Out-of-Range Numeric Values: Numeric variables exhibit several values outside their documented domains. Examples include ages exceeding the plausible range for secondary school students (e.g., 105, 170), studytime values as high as 11 despite the documented 1–4 encoding, and implausible grade values such as 132 (valid range: 0–20). These values represent clear schema violations and require conversion to missing values to prevent distortion of distributions and summary statistics.

Invalid or Inconsistent Category Encodings: Numerous categorical variables contain values do not present in their documented level sets. For example, the address variable should only contain 'U' (urban) or 'R' (rural), yet additional categories were observed. The reason attribute, documented to contain four valid categories, appears with five distinct values. Binary variables such as activities, romantic, and internet deviate from the documented 'yes'/'no' encoding through the presence of '0', suggesting data entry errors rather than alternative encoding standards. Since these values with 0 occur only once in each columns, they will be reviewed whether they represent true anomalies or broader formatting errors.

Identification of a Fully Corrupted Record: Row 649 displays multiple simultaneous schema violations: missing values in required fields (e.g., school, sex), extreme age values (170), invalid grade values (e.g., G1 = 142), and inconsistent binary encodings. The density of errors indicates wholesale record corruption rather than isolated typographical mistakes. Retaining this row would significantly distort descriptive statistics and modelling outputs; thus, it will be removed.

4. Data Cleaning

The goal of the cleaning process is to produce a dataset with consistent data types, standardised missing values, valid ranges, and properly encoded categorical variables. Each step below includes the justification for the action and its impact on data quality.

Remove corrupted row: The first step involved removing corrupted row 649, which showed multiple severe attribute violations. The density of errors indicates full-row corruption, not isolated typos. Retaining it would have distorted descriptive statistics, skewed distributional estimates, and potentially biased any model trained on the data; therefore, the row was removed entirely.

Standardize missing values: All categorical columns have non-standard placeholders ('-', 'X', 'nan' as a literal string), '0' value has been cleaned as they are concentrated in the deleted corrupted row. These are replaced with unified NaN so that missingness is explicitly recognized by pandas and preprocessing tools, avoiding false category inclusion and enable accurate imputation.

Correct typographical encoding errors: Age includes values such as '1O' and '2O', where the letter **O** replaces the digit 0. These are corrected to '10' and '20' so that the corrected values fall within realistic student age ranges and restores numeric validity and prevents conversion failures.

Assign correct data types

After replacing inconsistent type of values, columns are grouped as numeric, ordinal, or nominal and converted accordingly to match the UCI specification:

- **Numeric:** age, G1, G2, G3, absences
- **Ordinal:** Medu, Fedu, traveltime, studytime, failures, famrel, freetime, goout, health
- **Nominal:** school, reason, sex, address, famsize, Mjob, Fjob, activities, higher, internet, romantic

This ensures correct statistical handling of ordinal relationships, prevents inappropriate numeric interpretation of nominal variables.

Validate and standardise nominal categories: Whitespace is checked among nominal attributes and it is confirmed to be absent. Then category labels are corrected to match the UCI specification: "service" → "services". "at home" / "home" → "at_home", "others" → "other". This prevents artificial category fragmentation and improves interpretability of the results.

Apply domain-based validity rules: Domain-based validity rules were then enforced across the dataset. Values falling outside documented permissible ranges were replaced with NaN, as they represent structural errors rather than legitimate observations: failures (0-2), studytime (1-4), age (15-22), [G1, G2, G3] (0-20). Applying these domain constraints ensures that implausible or impossible values do not contaminate summary statistics or influence predictive modelling.

Convert Categorical Variables: To ensure efficient storage and enforce correct treatment of categories, all nominal variables were explicitly converted to unordered categorical types, and ordinal variables to ordered categorical types using pandas Categorical() function. This step guarantees that statistical procedures, encoders, and visualizations recognize the intended structure of each categorical field.

Review and retain statistical outliers: Finally, the dataset was examined for potential statistical outliers that, while extreme, remain plausible within the domain. High absenteeism values (20–35), grade extremes at both ends of the performance spectrum (0–5 and 18–20), and the upper-bound age of 22 were retained. These values represent genuine variation in student behavior and performance rather than errors and preserving them maintains the natural distributional richness of the dataset and avoids biasing models toward average cases.

Summary

In total, **1 corrupted row** was dropped and **52 rows** ($\approx 8\%$) were updated due to type corrections, invalid-value replacement, or category standardisation. After validation, the temporary `row_id` tracking column was removed.

As part of the data-cleaning workflow, all modifications were fully logged and exported to ensure transparency and reproducibility:

- **Dropped rows** were saved to **task1_dropped_rows.csv**, documenting the corrupted observations removed from the dataset.
- **All detected value-level changes** were recorded in **task1_value_changes.csv**, providing a complete audit trail of every correction applied.
- **Modified rows before cleaning** (52 rows) were extracted and saved as **task1_modified_rows_BEFORE.csv**, preserving the original erroneous values for reference.
- **Modified rows after cleaning** (52 rows) were saved as **task1_modified_rows_AFTER.csv**, showing the corrected versions following the cleaning process.

Task 2 Data Exploration

Task 2.1

Three variables were selected for visualization: **G2** (numeric), **Mjob** (nominal), and **studytime** (ordinal), based on their relevance to final grade student performance and the diversity of their data types as these metric help make visual analysis more informative and comprehensive.

Numeric Variable

A histogram is appropriate for continuous or discrete numeric variables because it shows the underlying distribution, central tendency, and potential skewness or outliers. This helps reveal performance patterns across students.

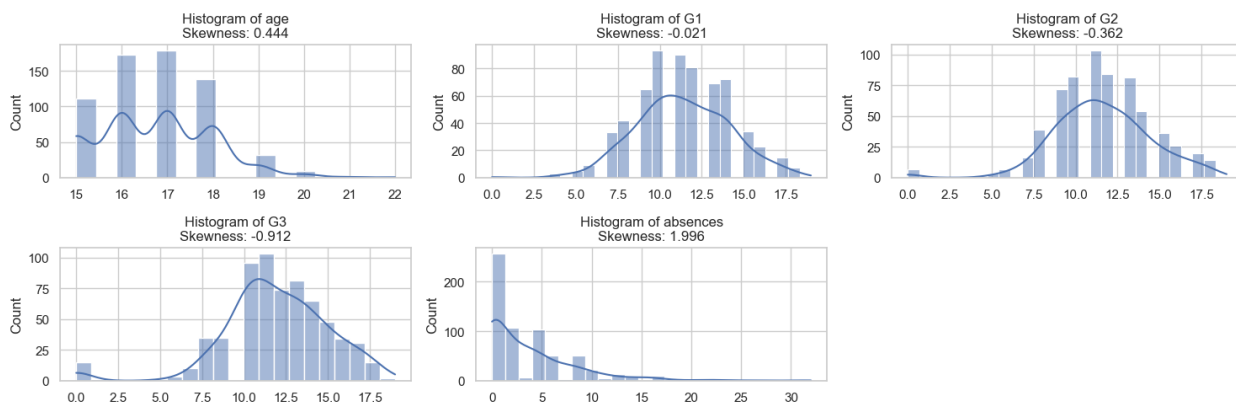


Figure 1. Histogram of all numeric variables

Among the numerical attributes (age, G1, G2, G3, absences), **G2** was chosen because it is the only variable with missing values and provides strong analytical value for comparing imputation methods in Task 3. **Absences** has no missing data, **G3** is the target and cannot be used, and **G1** is less informative than G2. G2 also has a clean,

moderately skewed distribution (-0.362), making it suitable for assessing how imputation affects variable shape. These factors make G2 the most appropriate numerical variable for this task.

The distribution of **G2 grades** is approximately bell-shaped, centred around scores of **10–13**, indicating that most students achieve mid-range performance. The slight left-skew is caused by a small number of very low grades (including zeros), while high grades (16–19) appear less frequently but remain within plausible limits. Overall, the distribution reflects typical academic performance patterns, with most students clustering around average grades and fewer achieving very low or very high marks.

Nominal Variable:

A countplot effectively displays the frequency of categories in a nominal variable. Since parental occupation is a non-ordered categorical feature, bar counts are the clearest way to compare category prevalence.

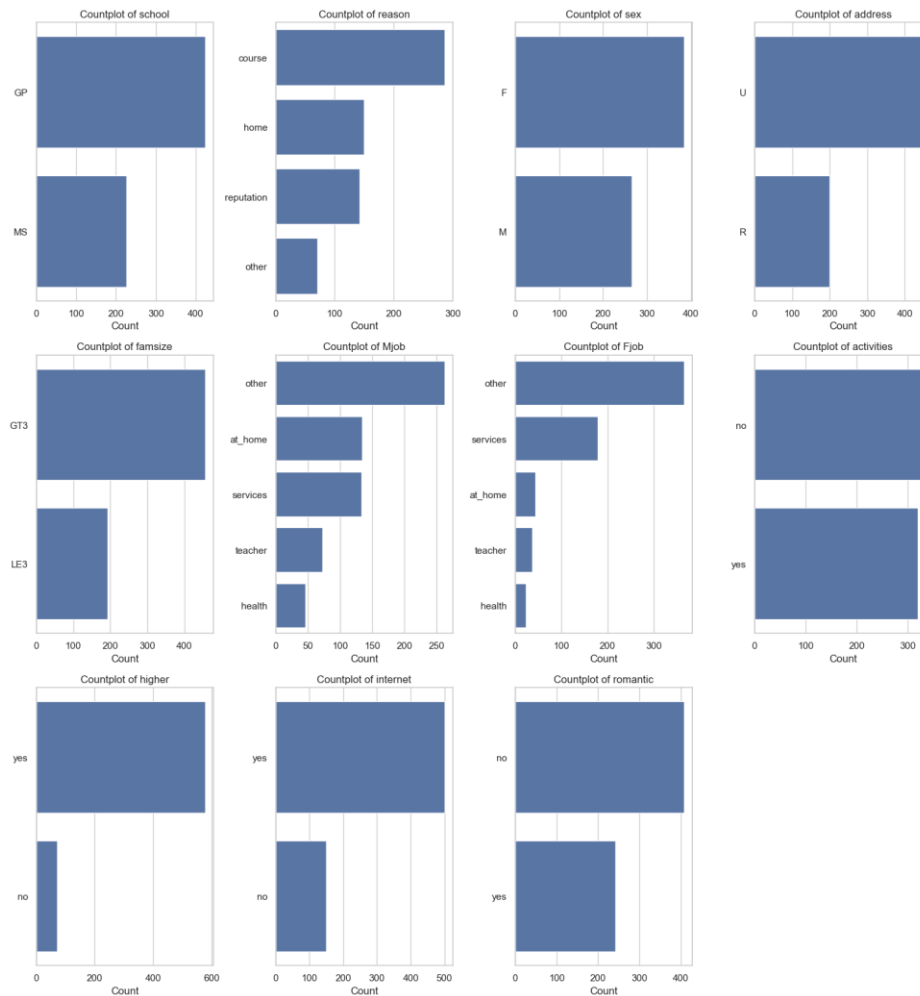


Figure 2. Countplot of all nominal variables

Mjob was chosen as the nominal variable because, unlike simpler two-category attributes (e.g., sex, address, activities), it has five meaningful. It is also more relevant to student performance than alternatives such as school, famsize, or lifestyle variables, as parental occupation is a well-established socio-economic factor influencing academic outcomes. This makes Mjob both more informative and more meaningful for visualization than other nominal features. The distribution of **mother’s occupation (Mjob)** is highly uneven, with the “other” category dominating the dataset. Occupations such as *services* and *at_home* appear at moderate frequencies, while *teacher* and especially *health* roles are much less common. This imbalance suggests substantial occupational diversity but

also indicates that certain professions are under-represented, which may influence any analysis that compares academic outcomes across parental job groups.

Ordinal variable

Although ordinal variables have an inherent order, they consist of discrete categories. A countplot highlights how students are distributed across study-time levels while preserving the natural ordering of the scale.

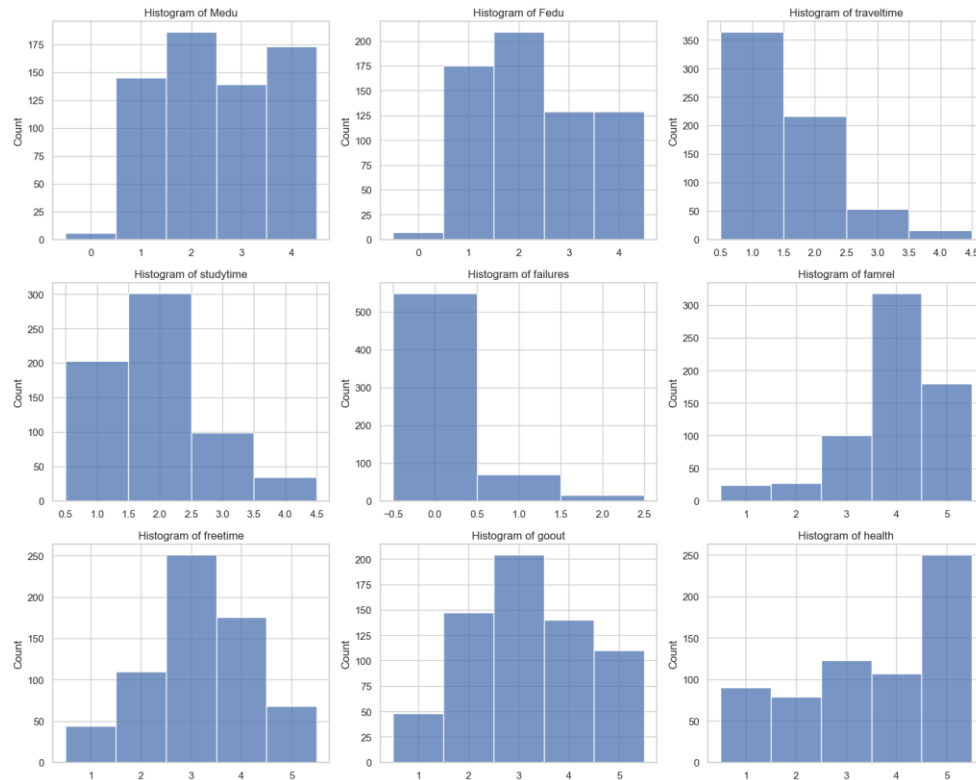


Figure 3. Countplot of all ordinal variables

For the ordinal variable, **studytime** was chosen because it directly reflects students' study behaviour, making it more meaningful for performance-related analysis than background or rating-based attributes. It also shows a balanced spread across its 1–4 scale, whereas many other ordinal variables are heavily skewed or concentrated in a single category. The distribution of *studytime* is strongly right-skewed, with most students reporting study levels of **1 or 2**, indicating relatively low weekly study commitment. Levels **3 and 4** are much less common, suggesting that only a small subset of students engage in higher-intensity study routines. This pattern reflects a typical school population where moderate study habits dominate, and extreme study behaviours are rare.

Task 2.2

Three pairs of variables were selected based on their hypothesis relevant to student performance and educational behavior.

Study Behavior Influences Academic Performance

Hypothesis: Students who spend more time studying tend to achieve higher final grades. Studytime reflects direct academic effort, while G3 is the final performance outcome. Their relationship is central to understanding learning behaviours. Box plot is chosen for visualization since it compares the distribution of a continuous response (G3) across ordered studytime categories while revealing differences in median, IQR, and distributional spread.

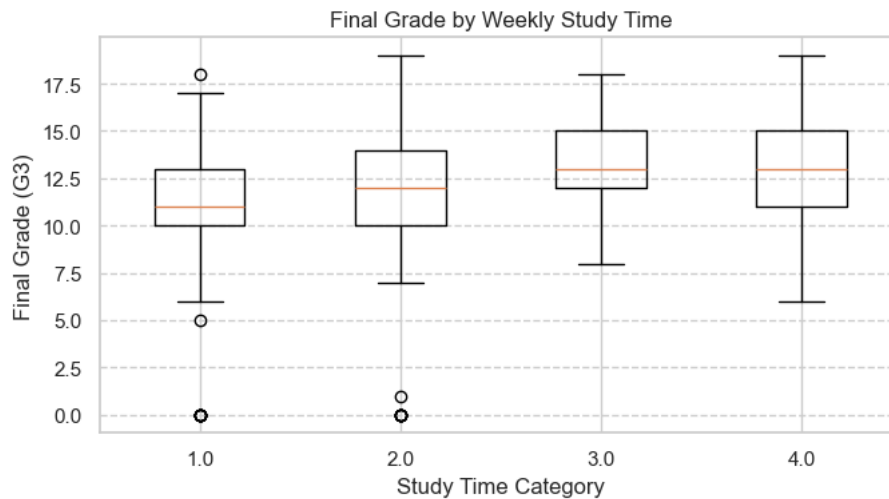


Figure 4. Boxplot of G3 by studytime category

The boxplot shows a slight upward trend in median G3 scores as studytime increases. However, extensive overlap in the interquartile ranges indicates considerable variability within each category. This suggests that although higher studytime may contribute positively to performance, it is not a dominant predictor and likely interacts with other academic or socioeconomic factors, providing weak support to the hypothesis.

Parental Education Is Associated With Student Achievement

Hypothesis: Students whose mothers have higher educational attainment achieve higher final grades because parental education is a dominant contribution to academic support, expectations, and learning environment quality. Boxplot is used to display how grade distributions shift across ordered parental education levels.

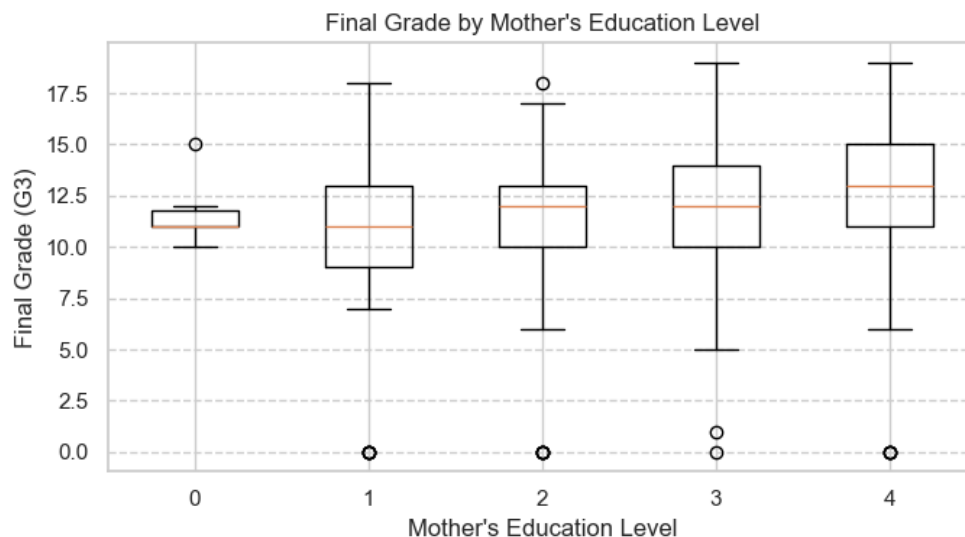


Figure 5. Boxplot of G3 by mother education category

The distribution of G3 scores shows a **gradual increase in median performance** as the mother's education level rises. However, the major overlap across levels 1 - 4 indicates that the relationship is **moderate**, and students from these education levels can achieve both high and low marks. Therefore, visualization suggests a moderate influence of maternal education on student achievement.

Internet Access Affects Study Behavior

Hypothesis: Students with home internet access study less due to increased availability of digital distractions. Countplot is used to show differences in the frequency distribution of ordered categories across nominal groups.

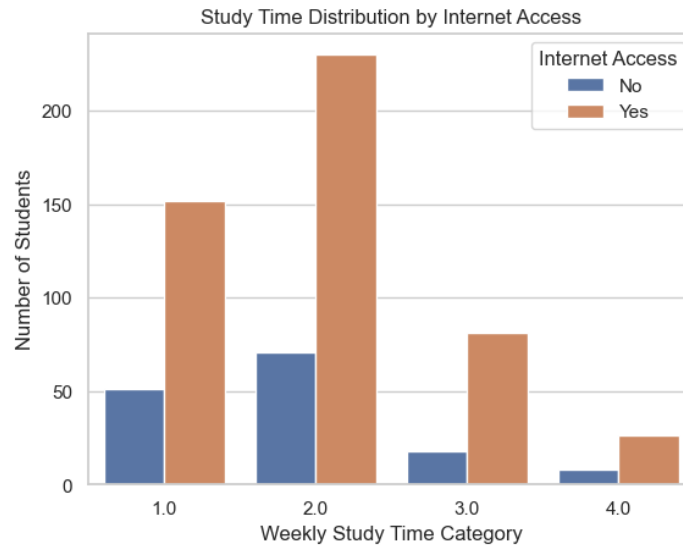


Figure 6. countplot of student studytime by internet access

The plot does **not** support the hypothesis. Students with internet access are not over-represented in lower studytime categories; instead, they appear at higher frequencies across all categories, including the highest (3–4). This suggests that internet access alone does not reduce study effort and that other factors (e.g., self-regulation, parental supervision) likely moderate the relationship.

Task 2.3

The scatter plot of all numeric variables in the dataset is visualized to inspect the overall relationships, trends, and noise patterns across all numeric features.

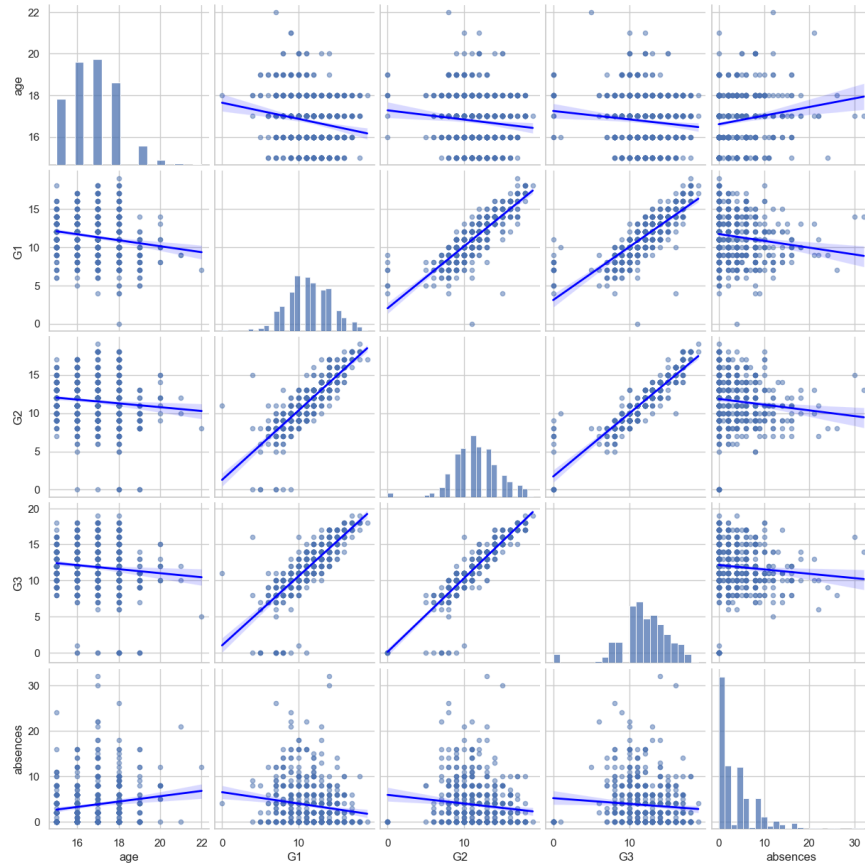


Figure 7. scatter matrix of all numeric variables

Based on the full scatter plot, both **age** and **absences** produce scatterplots that are visually cluttered, weakly correlated to other numeric features, making it difficult to produce a clear relationship with other columns. In contrast, all three grade pairs ($G1-G2$, $G2-G3$, and $G1-G3$) display **strong relationships to each other** and informative in predicting the academic assessment.

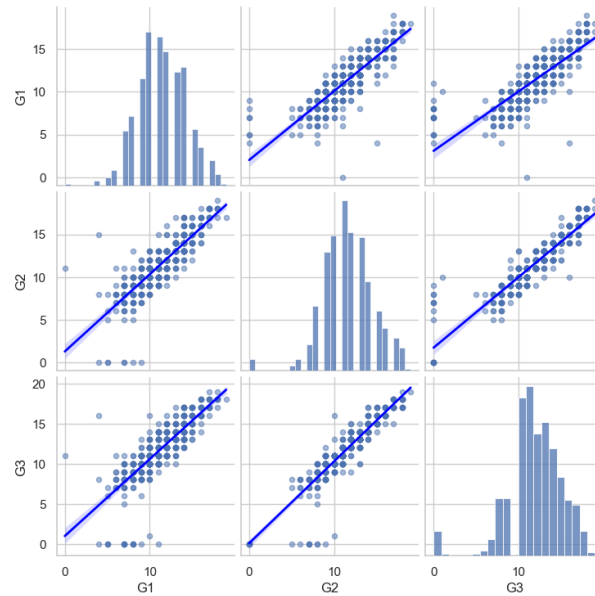


Figure 8. scatter matrix of $G1$, $G2$, $G3$

The scatter matrix of G1, G2, and G3 shows strong, clear positive linear relationships across all grade pairs, indicating that student performance is highly consistent throughout the academic year. G1 and G2 display the tightest clustering, demonstrating that early-term results closely predict mid-term achievement. The G2–G3 relationship is similarly strong, while the slightly wider spread in the G1–G3 plot reflects expected changes in performance over time. Overall, the visualization highlights that earlier grades are reliable predictors of final outcomes, making these three variables effective for illustrating academic progression.

Task 3 Analysis of Missing Values and Outliers

Task 3.1

The 3 imputation methods—mean, median, and linear interpolation—were chosen because they represent simple, widely used approaches that handle missing data in different ways.

These methods display plot of G2 (4 missing values) changes as it is inspected in Task 2.1. The plot will have a visualization comparison with the original dataset in Task 1 and mean, median, std, skewness changes are also investigated.

Median Imputation (imputed file at "student_fix2.csv")

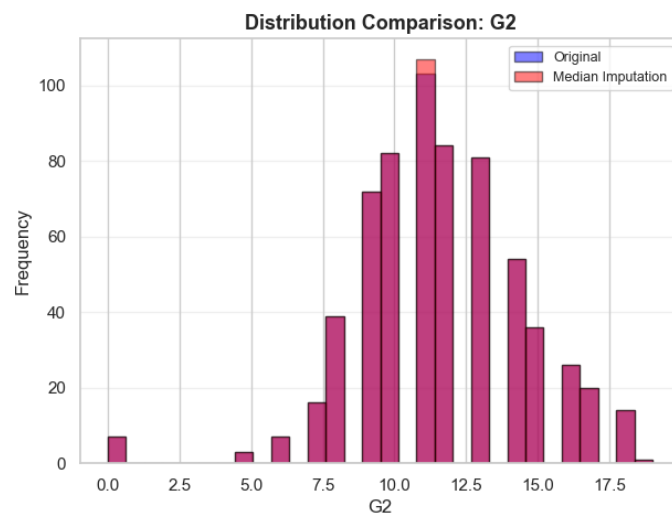


Figure 9. Comparison of median imputation and original G2 columns histogram

Median imputation introduces minimal visual changes to the G2 histogram, with only a slightly higher bar at the median category. The mean shifts by **0.03%**, the standard deviation decreases by **0.30%**, and skewness changes by **<1%**, while the median remains identical. Although median imputation is robust and preserves central tendency well, it **reduces variance** because repeated identical values replace genuine variability and can introduce **heaping** if many values are missing. This is suitable for G2 since it has few missing values.

Mean Imputation (imputed file at "student_fix1.csv")

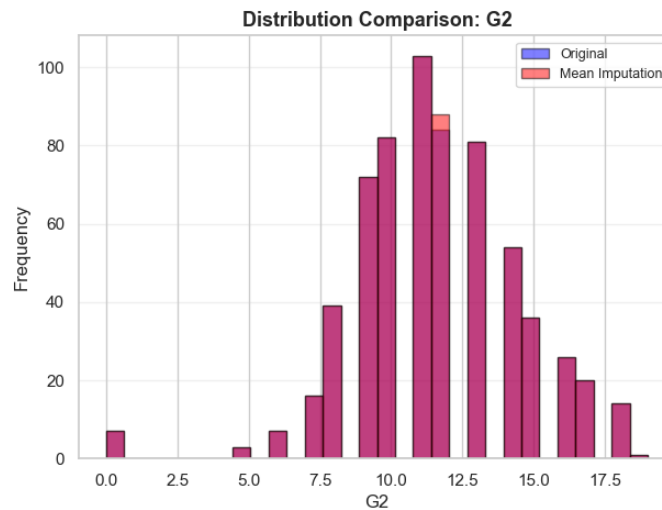


Figure 10. Comparison of median mean and original G2 columns histogram

Mean imputation produces similarly minimal changes, with a small increase in the bin around the mean. The standard deviation decreases by **0.30%**, and skewness changes by only **0.0011** (0.31%). Mean imputation inherently **underestimates variance** and can introduce **bias**, particularly when distributions are skewed or have outliers. In this dataset, the effect is negligible because the missing proportion is extremely small and the distribution of G2 is approximately symmetric. The distribution is preserved visually and numerically, but the bias limitation should be investigated if there are many missing values.

Linear Interpolation (interpolated file at "student_fix3.csv")

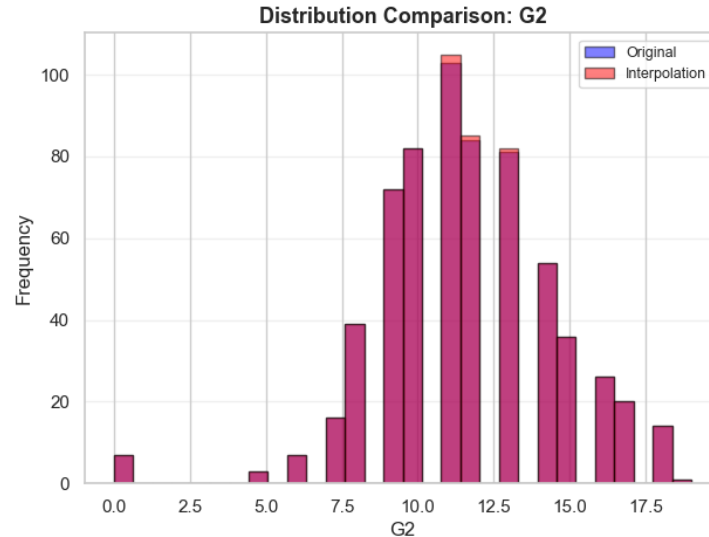


Figure 11. Comparison of median interpolation and original G2 columns histogram

The histogram shows that **interpolation imputation produces slight change** to the G2 distribution with small differences in the mid-range bins. The mean changes by **0.009%**, standard deviation decreases by **0.28%**, and skewness shifts by **<0.5%**; the median is unchanged. G2 sits in an ordered dataset (score listed in sequence), allowing local interpolation to produce realistic values. Unlike mean or median imputation, interpolation preserves natural variations because it uses neighboring values rather than a single central estimate.

Conclusion

Mean and median imputations slightly compress variance because they insert repeated values at fixed points, whereas interpolation maintains continuity by modelling local structure. These observations confirm that the statistical characteristics of G2—its center, spread, and shape—remain effectively intact under all three imputation strategies, with interpolation offering the most structure-preserving outcome.

Task 3.2

Do outliers affect standard deviation?

Standard deviation is **strongly affected by outliers** because it measures how far each value lies from the mean. Outliers are placed far from the center of the distribution, so they contribute disproportionately large squared deviations to the equation result, inflating the standard deviation even if only a single extreme value exists. As a result, the standard deviation may no longer reflect the variability of the typical data, and comparisons between variables or groups become misleading. This sensitivity is why robust alternatives (e.g., IQR, median absolute deviation) are often preferred when outliers are present, as they rely on medians and percentile-based ranges, which are not influenced by extreme values.

The same is applied in this dataset, outliers significantly inflate standard deviation. The grade variables increase by **23%** when outliers are included, and **absences** rises even more (**23.3%**). Even **age**, which has fewer extreme values, shows a small but noticeable increase (**1.55%**). This confirms that standard deviation in this dataset is strongly affected by outliers.

When should an outlier not be removed?

An outlier should not be removed when it represents a real, meaningful observation rather than an error. Outliers must remain when they reflect genuine extreme behavior (e.g., very high absences, unusually low grades), capture important subgroups, or signal rare but valid patterns in the dataset. Removing such values can erase critical insights, distort the distribution, reduce model validity, or eliminate groups that are academically or behaviorally at risk. Outliers should only be removed when they are proven to be data-entry errors, impossible values, or inconsistent with the variable's documented range.

Consider the numeric columns in the given file data, are there outliers in these columns of the given data set? How do you detect them?

In this dataset, all five numeric variables: age, absences, G1, G2, and G3, contain outliers, which fall into two categories: **domain-invalid outliers** and **statistical outliers**. Domain validation, based on UCI attribute definitions, identified impossible values such as ages **170** (outside 15 - 22), and grade values such as **132** (outside the 0–20 scale). These are errors and were removed or replaced with NaN.

Statistical outliers were detected using the IQR rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$). For the grade variables, scores at the lower and upper ends (0 - 5 and 18 - 20) appear as outliers but are legitimate extremes of academic performance. Absences above 20 also qualify as statistical outliers but plausibly reflect real behavioural patterns. Age 22, although rare, remains within the documented domain and should be retained.

In summary, every numeric column contains outliers, but only the **domain-invalid** values should be removed. The statistical outliers represent genuine variation and therefore remain important to preserve. Using both domain constraints and statistical detection techniques provides a balanced and defensible approach to maintaining data quality.

Task 4 External data

In this task performs a complete preparation workflow for the **MovieLens 100K dataset** using the `u.data` and `u.item` files.

1. Dataset Loading

The `u.item` and `u.data` files were loaded into separate dataframes, with column names applied according to the official MovieLens specification to ensure the correct interpretation of both the movie-level metadata and the user–movie rating interactions. Both dataframes use `item_id` as a common key. Because `u.data` contains multiple ratings per film while `u.item` provides a single metadata record for each title, several integrity checks were conducted prior to merging to prevent structural inconsistencies.

Completeness of key fields: A preliminary check confirmed that neither dataset contained missing values in the `item_id` column. This ensured that all records could be reliably matched during the merge.

1. **Uniqueness of records:** A second verification established that neither `u.item` nor `u.data` included duplicate rows. This is essential because duplicate metadata would misrepresent distinct films, and duplicate rating entries would incorrectly inflate user interaction counts.
2. **Referential integrity across datasets:** A third check verified that every `item_id` present in the ratings dataset appeared in the metadata file, confirming that no “orphan” ratings existed that would otherwise produce incomplete merge results.

With all integrity checks satisfied and the datasets proven to be consistent and fully compatible, the dataframes were merged using a **left join on `item_id`**. This approach preserves all rating interactions while augmenting them with the corresponding film attributes from the metadata file.

2. Initial Data Inspection

After merging, the combined dataframe was assessed using `info()`, `shape`, and `describe()` to evaluate its structure, completeness, and overall data quality.

- **Dataset Dimension:** The merged dataset contains **100,000 rows** and **27 columns**. All expected variables from both `u.data` and `u.item` are present, confirming a complete and accurate merge with no dropped or duplicated records.
- **Data type:** The current data types align with the UCI MovieLens specification: `release_date` is stored as a datetime object (e.g., *1-Jan-1999*), and `timestamp` as a single numeric value (e.g., *885853415*). While these representations are correct, they may introduce unnecessary complexity during modelling. Datetime strings and mixed temporal formats often behave like **categorical labels**, which can interfere with algorithms that expect strictly numerical inputs. Converting these fields into a consistent, numeric representation (such as *1-1-1999*) simplifies preprocessing and ensures that the temporal information can be interpreted meaningfully by machine learning models.
- **Missing values:** A targeted missing-value assessment identified such as 9 missing `release_date` entries, 100% missing `video_release_date`, and 13 missing entries of `IMDb_URL`. All other columns are fully complete with no missing values.
- **Data Validity:** All examined variables conform to the expected ranges and constraints defined by the MovieLens dataset. **Ratings** fall strictly within the 1–5 scale and **genre indicator columns** contain only binary values (0 or 1).

3. Data Quality Audit

Following the initial inspection, a comprehensive data quality audit was conducted to identify inconsistencies, formatting issues, and potential data integrity problems in the MovieLens dataset. The audit incorporated checks for formatting validity, missing values, range violations, and cross-field consistency.

Release Date Format Validation: The **release_date** field was evaluated for incorrect or inconsistent date formats, including invalid calendar dates and deviations from the expected DD-MMM-YYYY structure used in the MovieLens 100K dataset. All entries followed a valid and interpretable date format. No malformed or ambiguous dates were detected.

Whitespace Detection in Text Fields: Textual fields such as **title** and **IMDb_URL** were scanned for leading or trailing whitespace, which can introduce duplicate-like entries or hinder string matching. There are **15 movie titles** contained unnecessary leading or trailing whitespace and all other text fields were clean, exhibiting no extraneous whitespace. These whitespace issues represent formatting noise rather than semantic errors but should be corrected to ensure clean joins, consistent sorting, and accurate text analysis.

Year Mismatch Between Title and Release Date

For titles that embed a year (e.g., *Batman (1989)*), the numeric year in the title was compared with the year derived from the **release_date** field. While **55 mismatches** were, these discrepancies are not considered data errors. A title year may refer to the production year rather than release year, a re-release or remaster edition, or a stylistic title convention unrelated to official release chronology. Thus, these mismatches require no corrective action.

Data Range Validity Checks

Key fields were validated against the constraints defined by the MovieLens 100K documentation.

- **Ratings:** All values fall within the valid 1–5 integer rating scale.
- **Genre Indicators:** All genre columns contain only binary values (0 or 1), consistent with the dataset’s multi-genre encoding.
- **Timestamps:** All timestamps fall within the valid Unix epoch range with the earliest timestamp: **1997-09-20 03:05:10** and latest timestamp is **1998-04-22 23:10:38**

These values align with the temporal scope of the MovieLens 100K dataset, which collected ratings between late 1997 and early 1998.

Missing Values Review

A completeness assessment identified missing entries across several fields:

- **video_release_date:** **100% missing values** indicate that this field was never populated in MovieLens 100K.
- **release_date:** **9 missing values** and the rows with NaN **release_date** also have unknown movie titles. These cases may require targeted lookup or imputation, depending on analytical needs.
- **IMDb_URL:** There are **13 missing values** Missing URLs may affect external data linkage but do not compromise rating analyses.

Summary

The dataset demonstrates generally high structural and semantic integrity. Only minor formatting issues (whitespace in titles) and expected missing fields were observed. All range checks conform to the original MovieLens specification, and no critical data-quality failures were detected.

4. Data Cleaning

Strip Extra Whitespace from Titles: Leading and trailing whitespace in the title field was removed using `.str.strip()`. These inconsistencies do not change the meaning of the title but can disrupt merging, filtering, deduplication, and text-matching tasks. Cleaning them ensures consistent string formatting and improves the reliability of all text-based analysis and operations.

Standardize release_date to a Uniform Date Format: The `release_date` column was converted to a consistent and fully parsable date format. Standardization prevents parsing errors, ensures compatibility across libraries, and supports accurate extraction of temporal components such as year, month, and day. This enhances the integrity of any time-based analysis or feature engineering.

Convert timestamp to Human-Readable Datetime: The Unix timestamps attached to user ratings were transformed into conventional datetime format. This conversion makes temporal patterns in rating behaviour more interpretable, supports the creation of time-derived features (e.g., rating lag, weekday effects), and preserves all original temporal precision.

Remove video_release_date (100% Missing): The `video_release_date` column was dropped because it contains no usable data and is known to be intentionally unpopulated in MovieLens 100K. Removing a fully empty feature reduces noise, prevents unnecessary handling during analysis, and maintains a clean, compact dataset.

Retain Rows with Missing release_date: Rows lacking a release date were kept because they represent fewer than 0.02% of entries and contain valid rating information. Removing them would lead to avoidable loss of information. Since `release_date` is not required for core modelling tasks, retaining these rows preserves dataset completeness.

Retain Missing IMDb URLs: Missing values in `IMDb_URL` were left unchanged because the field represents optional external metadata and does not influence user–item interactions or predictive modelling. Keeping these missing entries avoids unnecessary imputation and maintains the dataset’s structural integrity.